

Amirmohammad (Amir) Mohammadi (“green card holder”)

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EDUCATION

Texas A&M University, College Station, Texas December 2026

Doctor of Philosophy in Computer Engineering

Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing

Sharif University of Technology, Tehran, Iran

February 2021

Master of Science in Electrical Engineering

Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring

PROFESSIONAL EXPERIENCE

Texas A&M University, College Station, Texas

September 2022 – December 2026

Graduate Research Assistant

- **Built** a few-shot whole slide image classification pipeline to investigate parameter-efficient fine-tuning of vision-language models for disease diagnosis in digital pathology applications.
- **Developed** a distribution-aware adaptation algorithm for Transformer-based foundation models as a parameter-efficient fine-tuning technique for waveform-based sensing applications.
- **Improved** the classification accuracy of a convolutional-based deep learning model by designing a time-frequency feature combination and engineering pipeline for audio-based applications.
- **Established** a cross-domain transfer learning benchmark showing that ImageNet-pretrained models outperform audio-pretrained models by 3 percentage points in acoustic classification.
- **Reduced** ground truth data requirements by a factor of 15 for physiological time-series signals by developing physics-informed neural networks for cuffless blood pressure measurement.
- **Disseminated** research findings through publications, presentations, and reports to funding agencies.

Sharif University of Technology, Tehran, Iran

July 2019 – February 2021

Graduate Student Researcher

- **Designed** an ECG + EDA electrical circuit for a wearable sensor (BLE SoC) with a 3× longer battery life compared to alternatives and delivered 94% mental stress detection accuracy across 18 participants.

SKILLS

- **Tools:** Python, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, MATLAB, C, Hugging Face, Transformers, Git, Linux, VS Code, Jupyter Notebook, High-performance Computing.
- **Interests:** Data Science, Applied AI, Foundation Models, Deep Learning Architectures & Techniques, Generative AI, Physics-Informed Machine Learning, Multi-Modal Learning, Efficient AI, AI Hardware.

AWARDS

Graduate research assistantship (over **\$250,000** in total funding)

September 2022 – December 2026

- Sponsors: Massachusetts Institute of Technology Lincoln Laboratory (**MIT LL**), Texas A&M University Department of Electrical & Computer Engineering (**TAMU ECEN**), National Institutes of Health (**NIH**).

Travel grants

Summer 2026

- Awarded by **IEEE Geoscience and Remote Sensing Society** to present at conference.

- Awarded by **TAMU Department of Electrical & Computer Engineering** to present at conference.

PEER-REVIEWED PUBLICATIONS

- Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification., **Mohammadi, A., Carreiro, D., Van Dine, A., & Peeples, J.**, (2026). IEEE International Geoscience and Remote Sensing Symposium. [\[link\]](#)
- Neighborhood Feature Pooling for Remote Sensing Image Classification., *Orvati Nia, F., Mohammadi, A., Al Kharsa, S., Naikare, P., Hampel-Aria, Z., & Peeples, J.*, (2026). Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. [\[link\]](#)
- Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks., **Mohammadi, A., Masabarakiza, I., Barnes, E., Carreiro, D., Van Dine, A., & Peeples, J.**, (2025). IEEE OCEANS. [\[link\]](#)
- Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models., **Mohammadi, A., Kelhe, T., Carreiro, D., Van Dine, A., & Peeples, J.**, (2025). IEEE OCEANS. [\[link\]](#)
- Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation., *Sel, K., Mohammadi, A., Pettigrew, R. I., & Jafari, R.*, (2023). Nature NPJ Digital Medicine, 6(1), 110. [\[link\]](#)
- An integrated human stress detection sensor using supervised algorithms., **Mohammadi, A., Fakharzadeh, M., & Baraeinejad, B.**, (2022). IEEE Sensors Journal, 22(8), 8216-8223. [\[link\]](#)

MANUSCRIPTS UNDER REVIEW

- Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification., *Ritu, J., Mohammadi, A., Carreiro, D., Van Dine, A., & Peeples, J.*, [\[link\]](#)
- Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification., **Mohammadi, A., Van Dine, A., & Peeples, J.**, [\[link\]](#)
- Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning in Few-Shot Vision-Language Modeling., [in review].

ACADEMIC SERVICE

- **Reviewer** for the IEEE International Conference on Acoustics, Speech and Signal Processing.
- **Reviewer** for the Expert Systems With Applications international journal.
- **Mentored** undergraduate and graduate students in designing & developing end-to-end AI projects.
- **Volunteered** for Night at the Zach during the Texas Science & Engineering Fair to showcase AI projects.
- **Served** as a session chair at the IEEE International Geoscience and Remote Sensing Symposium
- **Delivered** several lectures of “Machine Learning for Electrical Engineering” at Texas A&M University.

LANGUAGES

English (proficient); **Farsi** (native/bilingual); **Azerbaijani Turkish** (native/bilingual)